Exercise Sheet 4

Centrality and Structural Roles

5942UE Network Science

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4.A Centrality Measures

Exercise 4.A.1: Centrality by Hand

Consider the graph with nodes A, B, C, D, E, F and edges A–B, A–C, B–C, B–D, D–E, D–F, E–F.

1. Compute **degree centrality** $C_d(v) = \deg(v)/(|V| - 1)$ for all nodes.
2. Identify the shortest paths between all node pairs. Which node lies on the most shortest paths between other nodes? (Compute betweenness centrality by inspection.)
3. Which node has the highest degree centrality? Which has the highest betweenness centrality? Are they the same node?
4. Interpret: what structural role does the node with the highest betweenness play in this network?

Exercise 4.A.2: When Centrality Measures Disagree

Using `nx.karate_club_graph()`:

1. Compute degree, betweenness, closeness, and eigenvector centrality for all nodes.
2. Find the top-3 nodes for each measure. Do the top lists overlap?
3. Plot four subplots, each showing the network with node sizes proportional to one centrality measure.
4. Find a node that ranks **very differently** across measures. What does this reveal about its structural position?

4.B Structural Roles

Exercise 4.B.1: Embedded vs. Broker

Consider a graph with two triangles $1, 2, 3$ and $4, 5, 6$ connected by edge 3–4, plus node 7 connected only to nodes 3 and 4.

1. Is node 1 **embedded** or a **broker**? What evidence supports your answer?
2. Is node 7 **embedded** or a **broker**? What is its clustering coefficient C_7?
3. If edge 3–4 is removed, what happens to node 7's structural role?
4. Burt (1992) argues brokers gain "information advantage." What unique information could node 7 access that nodes in $1, 2, 3$ cannot?

Exercise 4.B.2: Betweenness as a Proxy for Brokerage

Using the two-triangle + node 7 graph from Exercise 4.B.1:

1. Build the graph in NetworkX. Compute betweenness centrality for all nodes.
2. Verify that node 7 has high betweenness despite $C_7 = 1$.
3. Explain the apparent contradiction: high clustering coefficient yet high betweenness.
4. On the karate club graph: find the 3 nodes with the highest betweenness but **lowest** clustering coefficient. What does this combination mean structurally?